Other Inorganic Elements

Copper: Functions with iron in the formation of haemoglobin.

Manganese: Has a similar effect though less marked than copper.

Cobalt: It is present in vitamin B_{12} which is also necessary for the formation of haemoglobin.

Zinc: Is found mainly in pancreatic tissue and may have an important part to play in the storage of insulin in the gland.

Clinical Significance of Minerals

1. **Calcium:** Deficiency of calcium in the body precipitates rickets in the children and osteomalacia in adults.
2. **Iron:** Reduced level or lack of iron causes anemia, certain hormonal changes in women.
3. **Iodine:** Lack of iodine in the body leads to enlargement of thyroid gland called as goiter.



Tetany (intermittent muscular spasm) is a condition caused by deficiency of calcium in the body.

STUDENT'S ACTIVITY

Word scramble: Find the deficiency disorders caused by nutrients

P	E	L	L	A	G	R	A	S	C
A	C	G	B	I	T	O	T	S	E
N	G	J	W	E	R	D	C	V	R
E	S	B	E	R	I	B	E	R	I
M	S	F	E	Q	E	T	E	Q	C
I	Z	E	R	T	I	O	G	G	K
A	X	L	M	G	U	F	E	F	E
B	C	N	Q	R	G	U	F	S	T
S	T	O	M	A	T	I	T	I	S
K	S	C	U	R	V	E	Y	A	T

STUDENT'S ACTIVITY

Prepare chart on various sources of nutrients.

VI. Water

Water is an important component with diet as it performs many vital functions in the body and hence is a part of balanced diet. Water makes up to 70% of total body weight inhuman beings. Water should be taken in enough quantities to prevent dehydration.

Distribution of Body Water

Total body water 40 litres (Average weight man (65-70kgs))

Extra cellular fluid 12 litres

Intra cellular fluid 28 litres



- Extra cellular fluid: Fluid is contained in the spaces outside the cell.
- Intra cellular fluid: fluid contained within the body cells.

Functions of Water

- Water is a major constituent of blood and tissue fluid
- It helps in transport of many substances from one compartment to another
- Provision of the moist internal environment required by all living cells
- Participation in all the chemical reactions occurring extracellularly and intracellularly
- Regulation of the body temperature

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